

Polarimetry in Hall A

E.Chudakov¹

¹Hall A, JLab

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Outline

1 Introduction

- Compton and Møller Polarimetry

2 Polarimeter in Hall A

- Compton Polarimeter
- Møller Polarimeter
- High field upgrade
- Atomic hydrogen trap

3 Summary

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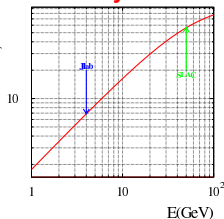
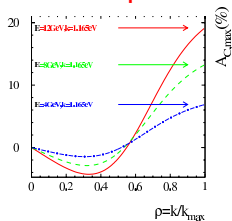
- Compton Polarimeter
- Møller Polarimeter
- High field upgrade
- Atomic hydrogen trap

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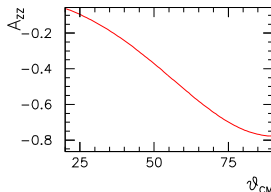
- Summary

Compton versus Møller Polarimetry

Compton Polarimetry



Møller Polarimetry



$$A = -\frac{7}{9}$$

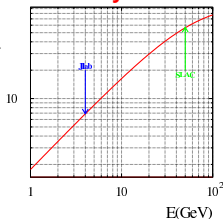
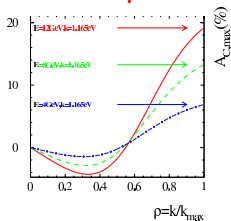
lab $\sim 180 \frac{mb}{ster}$

Kinematics/asymmetry

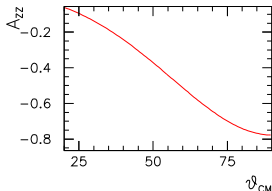
- Rad. corr. to Born $< 0.1\%$
- Detect γ at 0° , $e^- < E_{beam}$
- Strong $\frac{dA}{dk'}$ - need $\sigma E_\gamma / E_\gamma \ll 1$
- $A \propto kE$ at $E < 20 \text{ GeV}$
- $k' \sim 4\gamma^2 k$
- $T \propto 1/(\sigma \cdot A^2) \propto 1/k^2 \times 1/E^2$
- Rad. corr. to Born $< 0.3\%$
- Detect the e^- at $\theta_{CM} \sim 90^\circ$
- $\frac{dA}{d\theta_{CM}}|_{90^\circ} \sim 0$ - good systematics
- $A(E) \sim const$, $\frac{d\sigma}{d\Omega} \sim s^{-1}$
- Coincidence - no background

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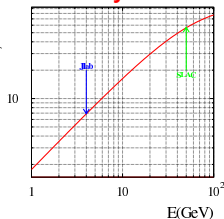
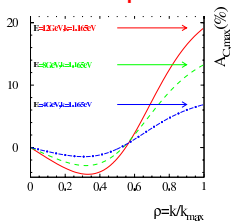
Polarized target

- $\mathcal{P}_{laser} \sim 100\%$
- Non-invasive measurement

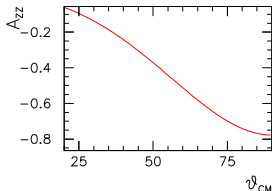
- Ferromagnetic target $\mathcal{P}_T \sim 8\%$
 - $> 1 \mu m$: invasive
 - Beam $I_B < 2 - 4 \mu A$ (heating)
 - Levchuk effect
 - Low $\mathcal{P}_T \Rightarrow$ dead time
 - Syst. error $\sigma(\mathcal{P}_T)$
 - $\sim 2\%$ for $B < 2 T$
 - $\sim 0.3\%$ for $B > 3 T$

Compton versus Møller Polarimetry

Compton Polarimetry



Møller Polarimetry



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$lab \sim 180 \frac{mb}{ster}$

Accuracy

Syst. error 3 → 50 GeV:

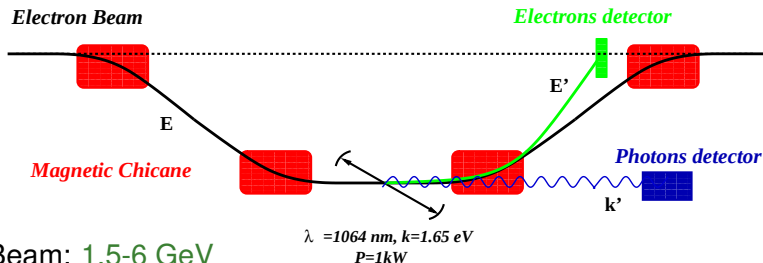
$\sim 1. \rightarrow 0.5\%$

Hard at < 1 GeV: (JLab project)

$\sim 0.8\%$

Syst. error $\sim 3\%$ typically,
0.5%(1%?) at high magn. fields

Compton Polarimeter at low energy: CW cavity



- Beam: 1.5-6 GeV
- Beam: 5 – 100 μA at 500 MHz
- Laser: 1064 nm, 0.24 W
- Fabry-Pérot cavity $\times 4000 \Rightarrow$ 1 kW
- Crossing angle 23 mrad
- e^- detector - Silicon μ -strip
- γ detector - calorimeter

Stat: 1.0% 30 min, 4.5 GeV, 40 μA

Syst: 1.2% at 4.5 GeV

Upgrade Plans - 1% at 0.85 GeV

- Laser: 532 nm, 0.1 W
- Cavity $\times 15000 \Rightarrow$ 1.5 kW
- Detector upgrade

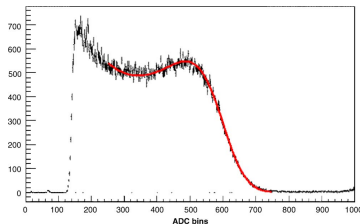
Electron and Photon Detection

Electron Detector

- 4 planes $\times$ 48 strips $\times$ 650 μm
- New: 192 strips $\times$ 250 μm
- Calibration for photon detector

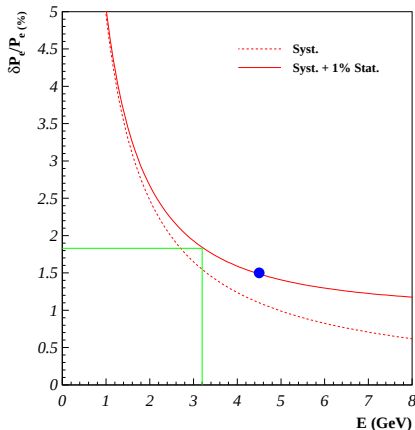
Photon Detector

- PbW, $2 \times 2 \times 23 \text{ cm}^3$
- 1 central crystal used
- New: one large GSO crystal



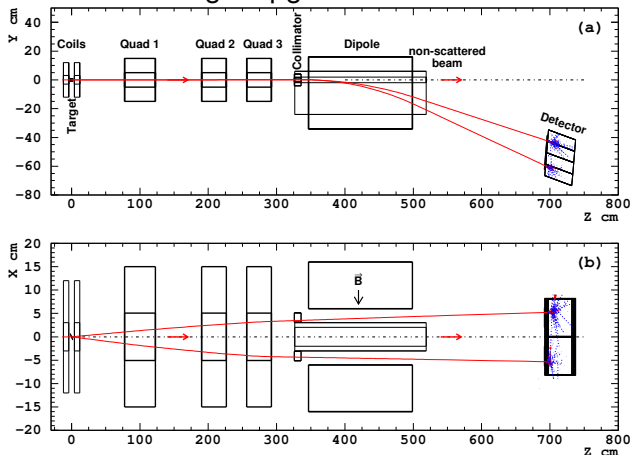
Existing Compton Polarimeter at 1064 nm

source	A error
Asymmetry	
Statistical	0.80%
Position and angle	0.30%
Background	0.05%
Dead time	0.10%
Cuts	0.10%
Light	
Polarization	0.50%
Analyzing power	
Response function	0.45%
Calibration	0.60%
Pile up	0.45%
Rad. correction	0.26%
Total systematic	1.15%
Total	1.40%



Hall A Møller Polarimeter

- 1998 - commissioned
- 2005 - target upgrade
- 2008 - target upgrade planned (high field)



- 0.8–6. GeV
- Minimal Levchuk
- $\sigma_{stat} = 1\%$ in $\sim 2-3$ min
- $B_z \sim 25$ mT field
- Foil at 20° to field
- Foils 5–30 μm thick
- Beam $< 2 \mu\text{A}$

Systematic Errors

The goal for the systematic error

Variable	Error		
	OLD	Present	PREX goal
Target polarization	3.5%	2.0%	0.5%
Target angle	0.5%	0.5%	0.0%
Analyzing power	0.3%	0.3%	0.3%
Levchuk effect	0.2%	0.2%	0.2%
Dead time	0.3%	0.3%	0.3%
Others	-	-	0.3%
Total	3.6%	2.1%	~1.0%

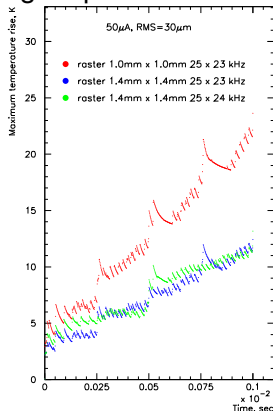
How to measure the “real” beam?

PREX will run at $50 \mu\text{A}$

Target heating $< 50^\circ\text{K}$ to avoid errors on depolarization

Run the injector as close to the regular running as possible

- Average current $\langle I_{beam} \rangle < 2 \mu\text{A} \Rightarrow \Delta T < 30 \text{ K}$
- Laser cycle: 1 ms pulse at 30 Hz
 $\Delta T_{pulse} \approx 12 \text{ K}$
- Beat frequency laser–chopper:
 $500 \text{ MHz} \Rightarrow 500/4 \text{ MHz}$
- Lower average rate $\Rightarrow 1\%$ statistically in 10–20 min



Appendix:Bunch suppression

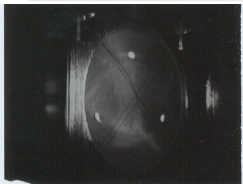
Options (from the draft of a paper by M.Poelker et al)

- G0: laser running at 499/16MHz - too long to install
- For regular bunch charges: laser at $\mathcal{F}_{laser} < \mathcal{F}_{RF}$
bunch suppression on the chopper.

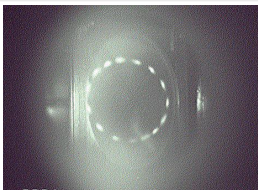
Beat frequency condition ($\mathcal{F}_{RF} = 499\text{MHz}$):

$$\mathcal{F}_{laser} \cdot (n + 1) = \mathcal{F}_{RF} \cdot n,$$

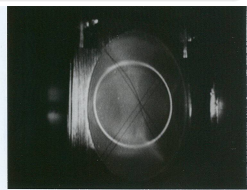
$n = 3, 4, 7, 15, 31, \dots$ - “magic” numbers



regular $\mathcal{F}_{laser} = \mathcal{F}_{RF}$



$n = 15$

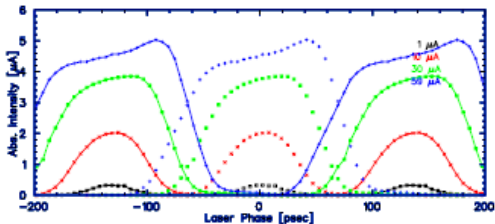


continuous

Appendix: Beat frequency mode - leak through

Pulses overlap

- $\tau_{pulse} \sim 200$ ps @ $50 \mu A$
- τ_{pulse} grows with $\mathcal{I}_{beam}$ (electro-repulsion)
- Fully open slit 110 ps
- No leak: $\Delta\tau > 160$ ps



Appendix: Optimization

- $n=15$ same slit $\Delta\tau = 133$ ps, contamination $\sim 5\%$ - bad
- $n=7$ same slit $\Delta\tau = 285$ ps, no contamination; other slit $\Delta\tau = 95$ ps leak $\sim 30\%$ - invasive for other halls
- $n=4$ other slit $\Delta\tau = 166$ ps non-invasive?

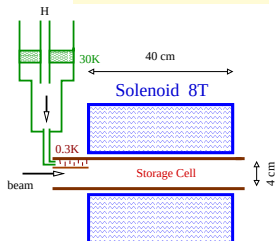
Macro-pulsing - tune beam

- Pulses $\Delta t > 4 \mu\text{s}$ at repetition rate $k \times 30 \text{ Hz}$
- Limitation: at $\mathcal{I}_{inst} = 50 \mu\text{A}$ accelerator stabilization time $\sim 100 \mu\text{s}$
- No micro-suppresssion: $\Delta t = 1 \text{ ms}$ at 30 Hz
- Micro-suppresssion $n=4$: $\Delta t = 1 \text{ ms}$ at 120 Hz

Polarized atomic hydrogen in a cold magnetic trap

E. Chudakov and V. Luppov, IEEE Trans. Nucl. Sci. **51**, 1533 (2004).

Ultra-cold traps



Atom H_1 : $\vec{\mu} \approx \vec{\mu}_e$, $E = -\vec{\mu}\vec{B}$

Population $\propto \exp(-E/kT)$

At 300 mK $P_e \sim 1 - 10^{-5}$

Density $\sim 3 \cdot 10^{15} \text{ cm}^{-3}$

Lifetime $> 1 \text{ h}$

Stat. 1% in 10 min at 100 μA

Contamination and Depolarization at 100 μA CEBAF

- Hydrogen molecules $< 2 \cdot 10^{-5}$
- Upper states $|c\rangle$ and $|d\rangle < 10^{-5}$
- Excited states $< 10^{-5}$
- Helium, residual gas $< 0.1\%$ - measurable
- Depolarization by beam RF $< 5 \cdot 10^{-5}$
- Ion, electron contamination $< 10^{-5}$
- Ionization heating $< 10^{-10}$

Expected depolarization $< 10^{-4}$

Limitations

- Problems $\propto I_b^2/\mathcal{F} \Rightarrow$ "continuous" beam
- Complexity of the target

Advantages

- Expected accuracy $< 0.5\%$
- Non-invasive, continuous, the same beam

Summary

Møller polarimetry in Hall A

- Old: $\sim 2.1\%$, using 8% Fe at ~ 25 mT , invasive
- New: $\sim 1\%$, using 8% Fe at ~ 4 T, invasive
- May provide $< 0.5\%$ accuracy with 100% polarized hydrogen, non-invasive, continuous

Compton polarimetry in Hall A

- Accuracy $\sim 1\%(4\text{GeV}) \Rightarrow 0.5\%(11\text{GeV})$, non-invasive